

Lab Final Project

Due Date : 20th May 2026

Computational Pipeline for Variant Detection using Simulated NGS Reads

1. Project Objective

This project provides hands-on experience with real bioinformatics workflows including:

- * Sequence analysis
- * Mutation introduction and analysis
- * NGS read simulation
- * Sequence alignment
- * Variant detection
- * Gene prediction
- * Functional interpretation
- * Network visualization

2. Project Overview

Each group will:

- * Work on a specific gene/protein
- * Apply assigned variation
- * Follow a common workflow
- * Perform both computational and biological analysis

3. Common Tasks (FOR ALL GROUPS)**

1. Obtain assigned gene sequence (FASTA format)
2. Introduce mutations (SNPs/insertions/deletions) as assigned
3. Compute edit distance between original and mutated sequence using python script.
4. Generate short reads (simulate NGS data by generating reads using a sliding window approach fixed read length with overlap).

5. Perform alignment using Galaxy (Bowtie tool)
6. Identify mismatches/variants from alignment output
7. Perform ORF/gene prediction
8. Analyze functional impact:

- * Coding vs non-coding
- * Synonymous vs non-synonymous

9. Create Cytoscape network:

Gene → Variant → Effect → Function

4. Group-wise Assignments

Group 1: SLC6A4 (Serotonin Transporter)

SNP Functional Classification

- Introduce only SNPs
- Classify:
 - Synonymous vs non-synonymous
- Translate protein and compare changes
- **Special Task:** Identify amino acid changes affecting polarity/charge

Position	Type
12	SNP
45	SNP
78	SNP

Group 2: KCNQ1 (Potassium Channel Gene)

Variation: Transition vs Transversion

- Introduce SNPs of both types
- Compare their frequency and effect
- **Special Task:** Determine which mutation type causes more protein change

Position	Type
60	SNP
150	SNP
300	SNP

Group 3: GJB2 (Gap Junction Protein)

Variation: Frameshift Mutations

- Introduce insertions/deletions
- Analyze frameshift impact
- **Special Task:** Identify premature stop codons

Position	Type
75	Insertion
180	Deletion

Group 4: PAH (Phenylalanine Hydroxylase)

Variation: Mixed Mutations

- Combine SNPs + indels
- **Special Task:** Compare ORF before vs after mutation

Position	Type
90	SNP
200	Insertion
350	Deletion

Group 5: TLR4 (Immune Receptor)

Variation: Real Variant Analysis

- Extract SNPs from dbSNP/ClinVar
- **Special Task:** Classify mutations as pathogenic/benign

Position	Type
120	SNP
250	SNP
400	SNP

Group 6: MTHFR (Metabolic Gene)

Variation: High Mutation Load

- Introduce 5–10 mutations
- **Special Task:** Analyze cumulative effect on protein stability (basic reasoning)

Position	Type
50	SNP
100	SNP
150	SNP
220	SNP
300	SNP
380	SNP

Group 7: CYP2C19 (Drug Metabolism)

Variation: Read Length Impact

- Simulate:
 - 50 bp reads
 - 150 bp reads
- **Special Task:** Compare alignment quality

Position	Type
70	SNP
160	SNP
280	SNP

Group 8: HIF1A (Hypoxia Gene)

Variation: Sequencing Error Simulation

- Introduce random errors in reads
- **Special Task:** Distinguish true mutations vs sequencing errors

Position	Type
55	SNP
140	SNP
260	SNP

Group 9: FOXP2 (Speech Gene)

Variation: Coding vs Non-Coding

- Analyze:
 - Coding region
 - Non-coding region
- **Special Task:** Compare functional impact

Position	Type
80	SNP
190	SNP
320	SNP

Group 10: PAX6 (Development Gene)

Variation: Gene Prediction Focus

- Perform ORF prediction
- **Special Task:** Compare predicted ORFs before and after mutation

Position	Type
65	SNP
170	Deletion
290	SNP

Group 11: LRRK2 (Neurodegeneration)

Variation: Alignment Performance

- Use Bowtie (Galaxy)
- **Special Task:** Compare alignment accuracy and mismatches

Position	Type
100	SNP
210	SNP
360	SNP

Group 12: NOD2 (Immune Gene)

Variation: Network Biology Focus

- Minimal mutation work
- Build detailed Cytoscape network:
 - Gene → Variant → Protein → Pathway
- **Special Task:** Add biological annotations (GO/pathways)

Position	Type
85	SNP
200	SNP
340	SNP

Important Instructions for ALL Groups

Each group MUST:

1. Follow complete pipeline:
 - Mutation → Edit Distance → Read Simulation → Alignment → Variant Detection → ORF → Interpretation
2. Submit:
 - FASTA (original + mutated)
 - Alignment output
 - Variant list
 - Cytoscape network
3. Answer clearly:

What is the biological significance of your mutations?

Note

Each group has a **unique variation**, so:

- Results must differ
- Analysis must be specific to assigned task
- Copying between groups will be penalized

11. Submission Guidelines**

Each group must submit the following components:

 ****A. Written Report (5–10 pages)****

****Structure (Mandatory)****

1. ****Title Page****

- * Project title
- * Group members
- * Assigned gene

2. ****Introduction****

- * Biological role of the gene
- * Relevance (disease / pathway / function)

3. ****Methodology****

- * Mutation introduction (with table)
- * Edit distance calculation
- * Read simulation method
- * Alignment (Bowtie parameters)
- * Variant detection approach
- * ORF prediction

4. ****Results****

- * **Mutation table**
- * **Alignment summary**
- * **Variant list**
- * **ORF comparison**

5. **Figures (Compulsory)**

- * **Alignment screenshot**
- * **Mutation visualization**
- * **ORF results**
- * **Cytoscape network**

6. **Discussion (VERY IMPORTANT)**

- * **Effect of mutations on:**
 - * **DNA sequence**
 - * **Protein sequence**
 - * **Gene function**
- * **Interpretation of results**

7. **Conclusion**

- * **Key findings**

**B. Presentation (10–12 Slides)**

**Required Slides**

- 1. Title slide**
- 2. Gene background**
- 3. Workflow diagram**
- 4. Mutation strategy**
- 5. Read simulation**
- 6. Alignment results**
- 7. Variant analysis**
- 8. ORF analysis**
- 9. Cytoscape network**

10. Biological interpretation

11. Conclusion

⚠ Rules:

- * Visual-heavy (less text)**
- * No reading from slides**
- * Max time: **10–12 minutes****

🧠 **C. Viva (Oral Examination)**

Each group member will be asked questions individually.

**Students must be able to explain:**

- * What is a mutation?**
- * Why did you choose those mutations?**
- * What is edit distance?**
- * How does Bowtie work (conceptually)?**
- * What is an ORF?**
- * What is the biological significance of your results?**

📁 **D. Files Submission (Compulsory)**

Submit a folder containing:

- * Original FASTA sequence**
- * Mutated FASTA sequence**
- * Simulated reads file**
- * Alignment output (SAM/BAM or screenshot)**
- * Variant table**
- * Cytoscape network file**

**13. Important Instructions**

- * All work must be **original****
- * Copying between groups will result in **zero marks****
- * Each member must contribute equally**
- * Late submission will be penalized**
